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INDIAN INSTITUTE OF INFORMATION TECHNOLOGY BHOPAL

Department of Computer Science & Engineering

End-Term Examination May - 2025

Semester:- 4th

Course : B.Tech

Code : CSE - 224

Branch: CSE Subject

Subject Name: Database Management System

Max. Marks:60

Time : 03hr

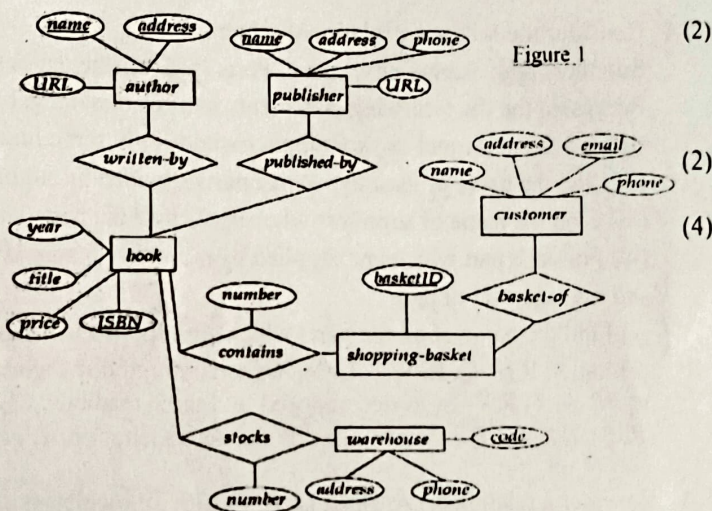
NOTE:

1. All the questions are compulsory.
2. While solving write question number and its part clearly.
3. Solve questions in given sequence and all the sub parts should be at same place.

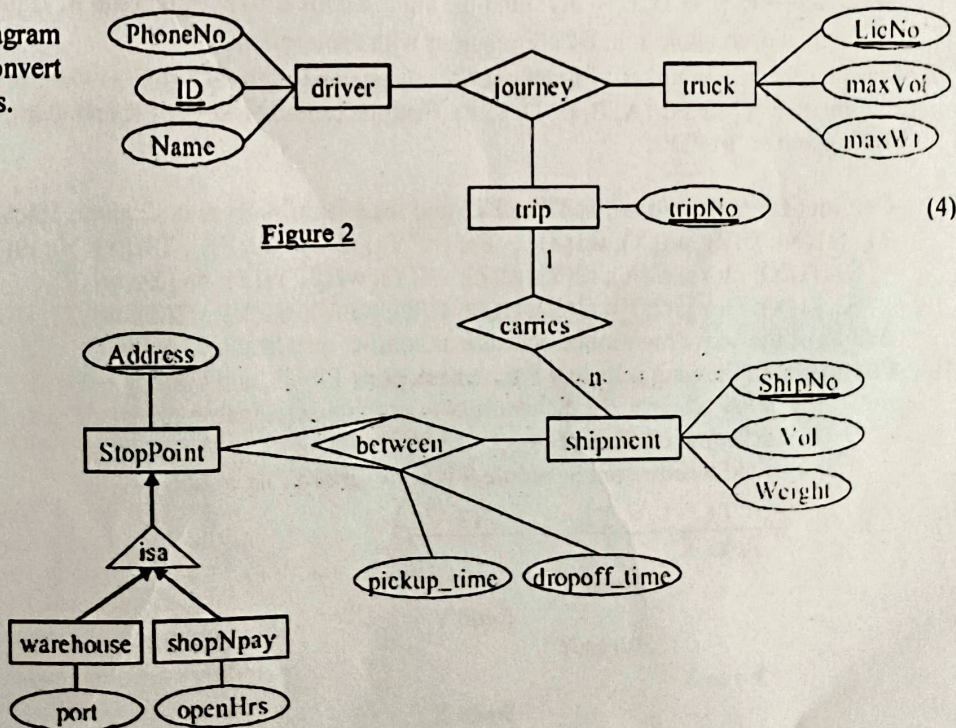
- 1.A (i) Discuss the terms "security of a database" and "integrity of a database" with the help of appropriate examples. (2)
(ii) Critically compare the relative advantages and disadvantages of a relational database vs hierarchical database. (2)
(iii) Highlight the importance of "foreign key" in design of databases. Explain with the help of an example. (2)
- B (i) Show that the axiom of transitivity is a particular case of pseudo transitivity axioms. Use Armstrong's axioms to prove the soundness of the union rules. (2)
(ii) Derive the division operation using the only suitable example. (2)

- 2.A Consider the E-R diagram given in figure 1 which models an online bookstore.

- (i) List the entity sets, their attributes and their primary keys.
- (ii) Suppose the bookstore adds music cassettes and compact disks to its collection. The same music item may be present in cassette or compact disk format, with differing prices. Extend the E-R diagram to model this addition, ignoring the effect on shopping baskets.



- B Consider the ER diagram given in figure 2 to convert it into relational tables.



3.A Consider the query:

*select * from r, s where upper(r.A) = upper(s.A);*

where "upper" is a function that returns its input argument with all lowercase letters replaced by the corresponding uppercase letters.

(i) Find out what plan is generated for this query on the database system you use.

(ii) Some database systems would use a (block) nested-loop join for this query, which can be very inefficient. How hash-join or merge-join can be used for this query. Discuss.

B Consider the relations shown below:

Student (Roll number, Name, Date_of_birth, Sex)

Course (Course number, Course_name, Instructor)

Enroll (Roll number, course number, Grade, Marks)

Represent the following statements into relational algebra expression and also into SQL query.

(i) Find the name of students who have got an 'A' grade in all the courses taught by Instructor "XYZ".

(ii) Find the name of courses in which more girls are enrolled than boys' students.

(iii) For each course in which more than 5 girls' students enrolled, find the course name, number of girls' students enrolled in the course and average marks obtained by them.

4.A Consider the following relational schema:

Suppliers (sid, sname, city, street) Parts (pid, pname, color) Catalog (sid, pid, cost)

Represent the the following statements into relational algebra and relational calculus expression.

(i) Find the name and city of supplier who supply some black and some blue parts.

(ii) Find the name of least expensive parts supplied by supplier 'ABC'.

(iii) Find the name of suppliers who supplies all the parts which are supplied by supplier 'XYZ'.

(iv) For each part which are supplied by more than 5 suppliers, find the name of part, number of suppliers and average cost of part.

(v) Find the name of all the parts which are supplied in "Indore" by supplier "ABC".

B. A relation R (P, Q, R, S, T, U, V, W) satisfying following set of functional dependencies $\{P \rightarrow QR, PU \rightarrow V, PR \rightarrow T, R \rightarrow S\}$ is decomposed into three relations. Check whether the decomposition $\{(P, Q, R, T), (R, S), (U, V, P)\}$ is lossless or not and dependency preserving or not.

5.A. Consider a relation R (A, B, C, D, E, F) with following set of functional dependencies holding on it: $\{AB \rightarrow C, AD \rightarrow E, B \rightarrow D, E \rightarrow A\}$. Find the highest normal form of relation R. If the relation is not up to BCNF then decompose it in BCNF relations with explanation.

B Consider the following set of functional dependencies $\{A \rightarrow BD, BC \rightarrow DE, B \rightarrow D, D \rightarrow AF\}$ hold on a relation schema R (A, B, C, D, E, F). Find the candidate keys of R and Compute the minimal cover of the given set of FD's.

6.A Consider the transactions T1, T2 and T3 and the schedules S1 and S2 given below.

T1: {r1(X); r1(Z); w1(X); w1(Z)} , T2: {r2(Y); r2(Z); w2(Z)} , T3: {r3(Y); r3(X); w3(Y)}

S1: r1(X); r3(Y); r3(X); r2(Y); r2(Z); w3(Y); w2(Z); r1(Z); w1(X); w1(Z)

S2: r1(X); r3(Y); r2(Y); r3(X); r1(Z); r2(Z); w3(Y); w1(X); w2(Z); w1(Z)

Which of the above mentioned schedule is conflict serializable?

B. Consider the following schedule S for transactions T1, T2, and T3:

i. Find whether the Schedule S is conflict serializable or not. If yes, find all possible serial schedule equivalent to S.

ii. Find whether the Schedule S is view serializable or not.

T1	T2	T3
Read X		
	Read Y	
		Read Y
	Write Y	
Write X		
		Write X
	Read X	
	Write X	